Arbitrarily many population maxima under uniform global dispersal

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Abstract

How does the total size of a population depend on how often its members move between habitats of different quality? We study a standard model in which individuals leave their location at a fixed rate and resettle anywhere in the habitat with equal probability, as larvae or seeds drawn from a well-mixed pool do. Bai and Li showed that, when habitat quality varies by less than a factor of about 1.6, total population has a single peak as the movement rate changes, and they asked whether several peaks are possible. We show that they are. A habitat with three quality levels can give two peaks; two levels never can; and suitably arranged habitats give any prescribed number of peaks. The mechanism is simple: a tiny patch of very rich habitat barely changes the average quality but adds a population bump at a movement rate comparable to its richness. The result is qualitative. In our explicit examples the extra peaks are very shallow, and the richest habitat is many times better than the poorest. We also give an exactly verified three-level example in which the best habitat is only about 14.2 times better than the worst, against 658 in our first example.

1 Introduction

A population spread over a patchy landscape is shaped by two forces. Local growth pulls each subpopulation towards what its own patch can support. Movement mixes the subpopulations together. A classical question in spatial ecology asks how the total equilibrium population depends on the rate of movement. Two facts are well known: with movement, the total can exceed the total resource [13, 7], and in two-patch models the total may rise and then fall as dispersal increases [9, 1, 10]. In the model studied here, Bai and Li [3] proved that the total has a single peak when habitat quality varies only modestly, and asked whether it can ever have more than one.

Whether there can be more than one peak matters in practice. If the response to connectivity has several peaks, a few experiments at one or two connectivity levels cannot be extrapolated, and the total population alone cannot identify the dispersal rate. We show that several peaks do occur, and that there is no upper limit on their number. We also report how large the effects are. They are mathematically robust but small, and we say so throughout.

Roadmap. Section 2 states the model and the results. Section 3 reduces the problem to a single scalar equation. Section 4 treats two and three resource levels, including the exact examples and their effect sizes. Section 5 constructs any number of peaks from rare rich patches, starting with an informal sketch. Section 6 gives an ecological reading. Section 7 places the work in the literature, and Section 8 lists open problems.

2 The model and the results

On the interval $\Omega = (0,1)$, consider

$$\partial_t u(x,t) = d\big(\overline{u}(t) - u(x,t)\big) + u(x,t)\big(m(x) - u(x,t)\big), \qquad \overline{u}(t) = \int_0^1 u(x,t)\,\mathrm{d}x. \tag{1}$$

Here $u(x,t)$ is the population density at location x. The *dispersal rate* $d > 0$ is the rate at which individuals leave their location. Emigrants enter a common pool and settle uniformly over the habitat, so the immigration rate at every point is $d\,\overline{u}$. The *resource* $m(x)$ is a bounded function bounded away from zero. Since the coefficient of u^2 is one everywhere, $m(x)$ is both the local intrinsic growth rate and the local carrying capacity. Dispersal costs nothing in this model.

The unique positive equilibrium is denoted by $\theta_{m,d}$, and the total equilibrium population by

$$M_m(d) = \int_0^1 \theta_{m,d}(x)\,\mathrm{d}x. \tag{2}$$

The habitat has measure one, so total population and mean density coincide. The curve $d \mapsto M_m(d)$ gives the total population as connectivity varies.

Equation (1) is the uniform nonlocal dispersal model (1.11) of Bai and Li [3]. The averaging operator was introduced by Nagylaki to represent global panmixia [14]. Bai and Li's Theorem 1.5 proves unimodality when $\operatorname{ess\,sup} m / \operatorname{ess\,inf} m < (1+\sqrt{5})/2 \approx 1.618$. Next to the question, they note that for perturbations of a constant the nonlocal model admits exactly one maximum. They then ask *whether it is possible for the total population to have multiple maximum points in the model (1.11)* [3, Section 5, p. 20]. We cite their arXiv preprint (version 1); we found no journal version. Their admissible class allows nonnegative nonconstant resources, so our positive examples lie inside it.

Theorem 1 (Arbitrarily many maxima from rare rich patches)**.** *Let $a > 0$ and let $N \geq 1$ be an integer. There is a sequence $m_j \in C^\infty([0,1])$ with the following properties.*

1. *For every j: $m_j > 0$, $m_j'(x) > 0$ for $0 \leq x \leq 1$, $\int_0^1 m_j(x)\,\mathrm{d}x = a$, and the map $d \mapsto M_{m_j}(d)$ has at least N distinct strict local maxima in $(0,\infty)$.*

2. *As $j \to \infty$, $m_j \to a$ in $L^p(0,1)$ for every $1 \leq p < \infty$, but not uniformly. The rich patches shrink in area but not in height. In our construction the contrast $\sup m_j / \inf m_j$ exceeds 2^{N-1} for all large j.*

The theorem should not be read as "nearly constant habitats can have many peaks". Item 2 says only that the heterogeneity occupies a vanishing area. The pointwise contrast does not shrink, and it grows without bound in N in our construction. The population gains between peaks are also small: they are of order ε, the area parameter of the rare patches. All this is consistent with Bai and Li's result for small pointwise contrast. Monotonicity of m_j is not a substantive constraint. Uniform mixing sees only the distribution of resource values (Section 3), so any arrangement, including an increasing one, gives the same curve. Before smoothing, $N+1$ resource levels suffice. Here $C^\infty([0,1])$ means the restriction of a smooth function defined on a neighbourhood of the closed interval.

Theorem 2 (An exact three-level example)**.** *Let*

$$p_1 = \frac{2405}{3504}, \qquad p_2 = \frac{1095}{3504} = \frac{5}{16}, \qquad p_3 = \frac{4}{3504} = \frac{1}{876},$$

and let m be the increasing step function

$$m = \tfrac{1}{20} \text{ on } (0,p_1), \qquad m = \tfrac{25}{12} \text{ on } (p_1, p_1+p_2), \qquad m = \tfrac{329}{10} \text{ on } (p_1+p_2, 1). \tag{3}$$

Then

$$M_m(3) = 1, \qquad M_m'(3) = 0, \qquad M_m''(3) = \frac{6042635}{1975192401} > 0, \tag{4}$$

and M_m has at least two strict local maxima. Every rearrangement of m (every resource with the same distribution of values) has the same population curve, and so satisfies (4) *and has the same two maxima. Moreover, every positive smooth strictly increasing resource that is sufficiently close in $L^1(0,1)$ to the increasing step function* (3)*, with values in a fixed compact subinterval of $(0,\infty)$, has a nondegenerate local minimum near $d = 3$ and at least two distinct strict local maxima. For such approximations the exact identities* (4) *are not claimed: the location and value of the minimum move with the resource.*

Theorem 3 (Two resource levels)**.** *Suppose m takes the distinct values $a < b$, on sets of measures p and $q = 1 - p$, where $a > 0$ and $0 < p < 1$. Then*

$$\operatorname{sgn} M_m'(d) = \operatorname{sgn}(a + b - 2d). \tag{5}$$

The unique local and global maximum occurs at $d = (a + b)/2$, and

$$\max_{d>0} M_m(d) = \frac{pa + qb + \sqrt{pa^2 + qb^2}}{2}. \tag{6}$$

Among resources taking finitely many values, three is therefore the smallest number of levels that permits multiple strict local maxima.

Proposition 4 (A low-contrast three-level example)**.** *Let $d_0 = 76/25$ and*

$$\theta_1 = 1 - 10^{-6}, \qquad \theta_2 = 1 + 10^{-7}, \qquad \theta_3 = \tfrac{57}{5}, \qquad m_i = \theta_i + d_0 - \frac{d_0}{\theta_i}.$$

The linear system $\sum_i p_i = 1$, $\sum_i p_i\theta_i = 1$, $\sum_i p_i(1-\theta_i)/(\theta_i + d_0/\theta_i) = 0$ has a unique solution, and it is positive. The three-level resource with values m_i on sets of measures p_i satisfies $M(d_0) = 1$, $M'(d_0) = 0$ and $M''(d_0) > 0$. It therefore has at least two strict local maxima. Its contrast is

$$\frac{m_3}{m_1} = \frac{14173319160000}{999994960001} \approx 14.1734.$$

So the contrast needed for multiple maxima among three-level resources lies between Bai and Li's 1.618 and the certified 14.1734. Section 8 records numerical evidence that 14.17 is close to optimal for three levels, and the corresponding open problem. We also show (Corollary 9) that every critical dispersal rate lies strictly between the smallest and largest resource values.

3 Scalar reduction and parameter dependence

Uniform global mixing makes the equilibrium population depend only on the probability distribution μ of the resource values, not on their spatial arrangement. Write z for a resource value and let

$$0 < \ell \le z \le b < \infty \quad \mu\text{-almost everywhere.}$$

Expectations in this section are with respect to μ. At equilibrium, each location balances local growth against the net flow $d(M - \theta)$ from the common pool:

$$d(M - \theta) + \theta(z - \theta) = 0, \qquad M = \mathbb{E}\theta. \tag{7}$$

For a given pool level M, solving the quadratic gives the local density

$$\theta = \Phi(z, d, M) := \frac{z - d + \sqrt{(z-d)^2 + 4dM}}{2}, \qquad M = \mathbb{E}\Phi(z, d, M). \tag{8}$$

We call the second equation the *closure*. It says that the pool level must be reproduced by averaging the local densities it induces.

Lemma 5 (Existence, uniqueness and analytic dependence)**.** *For every $d > 0$, equation* (8) *has a unique positive solution $M = M_\mu(d)$. It satisfies*

$$\ell \le M \le b, \qquad \ell \le \Phi(z, d, M) \le b.$$

The function M_μ is real analytic on $(0, \infty)$. If

$$A = 2\theta + d - z = \theta + \frac{dM}{\theta}, \qquad D = 1 - d\mathbb{E}\frac{1}{A}, \tag{9}$$

then

$$D = \frac{1}{M}\mathbb{E}\frac{\theta^2}{A} > 0. \tag{10}$$

In words, A is the local stiffness: $1/A$ measures how strongly the density at a site responds to a change in its inflow. The quantity D is the stabilising factor of the closure. It is the derivative of $M - \mathbb{E}\Phi$ with respect to M, and its positivity drives every implicit-function step below.

Proof. For fixed z, d, differentiating (8) in M gives $\Phi_M = d/A$. The identity

$$\Phi - M\Phi_M = \frac{\Phi^2}{A} > 0$$

shows that $\mathbb{E}\Phi(z, d, M)/M$ is strictly decreasing in $M > 0$. At $M = \ell$, every integrand $\Phi(z, d, \ell)$ is at least ℓ, whereas at $M = b$ every integrand $\Phi(z, d, b)$ is at most b. Continuity therefore gives a root in $[\ell, b]$; strict decrease of the quotient gives uniqueness among all positive roots. The density bounds follow because the positive root of the quadratic lies between z and M.

Averaging the displayed identity at a root proves (10). The map $F(d, M) = \mathbb{E}\Phi(z, d, M) - M$ is real analytic near every $(d, M) \in (0, \infty)^2$. Indeed, the discriminant is bounded away from zero on a sufficiently small common neighbourhood, uniformly for $z \in [\ell, b]$, and the resulting convergent power series can be integrated term by term. At a root, $F_M = -D < 0$. The analytic implicit function theorem proves the assertion. □

Direct substitution into (7) gives, for $c > 0$, the *scaling law*

$$M_{cm}(d) = c\, M_m(d/c). \tag{11}$$

Only the ratio of dispersal rate to resource level matters. Multiplying all resources by c moves every peak from d_* to cd_* and leaves the number and strictness of peaks unchanged.

Lemma 6 (Population excess and endpoint limits)**.** *Let $\bar{m} = \mathbb{E}z$. If μ is not concentrated at one point, then*

$$M_\mu(d) > \bar{m} \quad (d > 0), \qquad \lim_{d \downarrow 0} M_\mu(d) = \lim_{d \to \infty} M_\mu(d) = \bar{m}. \tag{12}$$

For every resource distribution,

$$M_\mu(d) \le \frac{\mathbb{E}z + \sqrt{\mathbb{E}z^2}}{2}. \tag{13}$$

The inequality $M_\mu > \bar{m}$ is the phenomenon in which total population exceeds total carrying capacity [13, 7]. For this model it holds at every dispersal rate.

Proof. Dividing (7) by θ and averaging yields

$$M - \bar{m} = d\left(M\mathbb{E}\frac{1}{\theta} - 1\right) \geq 0,$$

by Jensen's inequality. The inequality is strict unless θ is constant almost surely; the equilibrium equation then makes z constant as well. The small-d limit follows uniformly from (8) and $\ell > 0$. For the other limit, the uniform density bounds and (7) give $\theta - M = O(d^{-1})$. Averaging (7) gives $\mathbb{E}\theta(z-\theta) = 0$, and thus $M(\bar{m}-M) = O(d^{-1})$. Since $M \geq \bar{m} > 0$, the limit follows. Finally, $\mathbb{E}(\theta - z/2)^2 = \mathbb{E}z^2/4$ by the same identity. Cauchy–Schwarz, applied to the mean of $\theta - z/2$, proves (13). □

Lemma 7 (Continuity under resource approximation)**.** *Let $m_k, m : (0,1) \to [\ell, b]$, where $0 < \ell < b < \infty$, and suppose $\|m_k - m\|_{L^1} \to 0$. On every compact interval $I = [d_0, d_1] \subset (0, \infty)$,*

$$M_{m_k} \longrightarrow M_m \quad \text{in } C^2(I).$$

In particular, any finite collection of strict inequalities between population values at prescribed dispersal rates in I is preserved for sufficiently large k. So are strict sign conditions on M' and M'' on I.

The full C^2 statement is used only in the smoothing step of Theorem 2. The other applications need only the preservation of strict inequalities between values.

Proof. The root function obeys

$$0 < \Phi_z = \frac{\Phi}{2\Phi + d - z} < 1.$$

Consequently $|\mathbb{E}_{m_k}\Phi - \mathbb{E}_m\Phi| \leq \|m_k - m\|_{L^1}$ for every (d, M): the closure functions converge uniformly in (d, M). All their positive roots belong to $[\ell, b]$. Compactness, continuity and the uniqueness in Lemma 5 imply uniform convergence of the roots for $d \in I$. Otherwise a sequence $d_k \to d_*$ and a convergent subsequence of roots would give a second positive root at d_*.

On the compact set $K = I \times [\ell, b]$, the derivatives of Φ in (d, M) through order three are smooth functions of $z \in [\ell, b]$, with uniformly bounded z-derivatives. Their expectations under m_k therefore converge to those under m, uniformly on K. Using $\ell \leq \theta, M \leq b$ and $A = \theta + dM/\theta \leq b + d_1 b/\ell$, we get the uniform lower bound

$$D = \frac{1}{M}\mathbb{E}\frac{\theta^2}{A} \;\geq\; \frac{\ell^2}{b\,(b + d_1 b/\ell)} > 0 \qquad (d \in I), \tag{14}$$

for every resource with values in $[\ell, b]$. The implicit formulas $M' = \mathbb{E}[\Phi_d]/D$ and $M'' = (\mathbb{E}[\Phi_{dd} + 2\Phi_{dM}M' + \Phi_{MM}(M')^2])/D$, where $D = 1 - \mathbb{E}\Phi_M$, are continuous in these expectations and in $1/D$. This gives the asserted C^2 convergence. □

Remark 8 (Strictness from analyticity). If $0 < d_- < d_0 < d_+$ and

$$M(d_0) > \max\{M(d_-), M(d_+)\},$$

then M attains a maximum on $[d_-, d_+]$ at an interior point. The trap forces M to be nonconstant on (d_-, d_+), so by analyticity it is nonconstant on $(0, \infty)$. A nonconstant real analytic function has only isolated critical points, so this maximum is strict. Traps whose open intervals (d_-, d_+) are disjoint therefore give distinct strict local maxima, even when two traps share an endpoint. This lets us work with finitely many population values, without a numerical root count for M'.

4 Two and three resource levels

Differentiating (7) gives

$$A\theta' = M - \theta + dM', \qquad M' = \frac{\mathbb{E}[(M-\theta)/A]}{D}. \tag{15}$$

The numerator has a useful pairwise representation. For independent resource values $z, \widetilde{z}$, put $\theta = \Phi(z, d, M)$, $\widetilde{\theta} = \Phi(\widetilde{z}, d, M)$ and define $\widetilde{A}$ correspondingly. Then

$$\begin{aligned}\mathbb{E}\frac{M-\theta}{A} &= \frac{1}{2}\mathbb{E}\left[\frac{(\theta-\widetilde{\theta})^2}{A\widetilde{A}}\left(1 - \frac{dM}{\theta\widetilde{\theta}}\right)\right] \\ &= \frac{1}{2}\mathbb{E}\left[\frac{(\theta-\widetilde{\theta})^2(z+\widetilde{z}-2d)}{(\theta+\widetilde{\theta})A\widetilde{A}}\right].\end{aligned} \tag{16}$$

The first equality is the standard two-copy expression for a covariance. The second uses

$$z + \widetilde{z} - 2d = (\theta + \widetilde{\theta})\left(1 - \frac{dM}{\theta\widetilde{\theta}}\right).$$

Each pair of resource values pushes the population up when the dispersal rate is below their average, and down when it is above.

Proof of Theorem 3. Let $x < y$ be the equilibrium densities at resources $a < b$. Formula (16) reduces to

$$\mathbb{E}\frac{M-\theta}{A} = \frac{pq(y-x)^2(a+b-2d)}{(x+y)A_xA_y}.$$

Every factor except $a + b - 2d$ is positive. This proves (5). At the unique critical rate $d_* = (a+b)/2$, the pairwise identity gives $d_*M = xy$. Subtracting the two resource equations then gives $y - x = (b-a)/2 =: \Delta$. The relations $M = x + q\Delta$ and $d_*M = x(x+\Delta)$ give $x^2 - ax - d_*q\Delta = 0$, and $a^2 + 4d_*q\Delta = pa^2 + qb^2$, so

$$x = \frac{a + \sqrt{pa^2 + qb^2}}{2}.$$

Adding $q\Delta$ proves (6). A one-level resource gives a constant curve, and Theorem 2 gives two maxima with three levels. This proves the minimality statement for finitely-valued resources. □

Corollary 9 (Each fixed resource has finitely many critical rates)**.** *Let a positive bounded nonconstant resource have essential infimum $\ell > 0$ and essential supremum b. Then $M'(d) > 0$ for $0 < d \le \ell$ and $M'(d) < 0$ for $d \ge b$. Its critical rates form a finite subset of (ℓ, b).*

Proof. The signs follow directly from (16). Nonconstancy gives strictness even at the two endpoints. Analyticity and the nonzero endpoint signs exclude infinitely many zeros of M' in the compact interval $[\ell, b]$. □

At a stationary population value, differentiating the first equation in (15) once more, using $A' = 2\theta' + 1$ and $\mathbb{E}\theta'' = M''$, yields

$$DM'' = -2\mathbb{E}\frac{(\theta')^2 + \theta'}{A}, \qquad \theta' = \frac{M-\theta}{A}. \tag{17}$$

How the examples are found. The weights in Theorem 2 look arbitrary, but they are easy to construct. Fix d and $M = 1$ and choose the equilibrium densities θ_i first. Equation (7) then gives the resources, $m_i = \theta_i + d - dM/\theta_i$. The three conditions $\sum p_i = 1$, $\sum p_i\theta_i = 1$ and $M'(d) = 0$ are linear in the habitat fractions p_i. It remains to check that the p_i are positive and that (17) gives $M'' > 0$. Theorem 2 uses $d = 3$ and $\theta = (4/5, 4/3, 30)$.

Proof of Theorem 2. At $d = 3$, the equilibrium values, their A-values, and their derivatives at a stationary mean are

θ_i	A_i	θ_i'
$4/5$	$91/20$	$4/91$
$4/3$	$43/12$	$-4/43$
30	$301/10$	$-290/301$

Indeed $\sum_i p_i\theta_i = 1$, and $m_i = \theta_i + 3 - 3/\theta_i$, which verifies all equilibrium equations exactly. Also,

$$\sum_i \frac{p_i}{A_i} = \frac{5}{21}, \qquad \sum_i p_i \frac{1-\theta_i}{A_i} = 0.$$

Thus $D = 2/7$ and $M'(3) = 0$. The remaining exact expectation is

$$\sum_i p_i \frac{(\theta_i')^2 + \theta_i'}{A_i} = -\frac{6042635}{13826346807}.$$

Formula (17) gives (4). Since M depends only on the distribution of values, the same holds for every rearrangement of m.

Two maxima. Since $d = 3$ is a strict local minimum, there are points $d_- < 3 < d_+$ arbitrarily close to 3 with $M(d_\pm) > M(3) = 1$. By (12), $M(d) \to \bar{m} = 347/480 < 1$ as $d \downarrow 0$ and as $d \to \infty$. Hence M attains a maximum on $(0, 3)$ and another on $(3, \infty)$, each at an interior point. The traps $(\alpha, d_-, 3)$ and $(3, d_+, \gamma)$, for suitable $\alpha < d_-$ and $\gamma > d_+$, share the endpoint 3 but have disjoint open intervals, so the two maxima are distinct. Both are strict by Remark 8.

Smooth increasing versions. Extend the increasing step function m of (3) constantly beyond the endpoints, convolve it with a nonnegative smooth mollifier, and add κx with $\kappa \downarrow 0$. The resulting functions are positive, smooth and strictly increasing, take values in $[1/20, 329/10 + 1]$, and converge to m in L^1. (Spatial L^1 approximation by increasing functions requires this increasing arrangement; it is not possible for an arbitrary arrangement.) Choose a closed interval $I \ni 3$ on which $M'' > 0$, with M' negative at its left endpoint and positive at its right endpoint. Lemma 7 gives C^2 convergence on I, so every sufficiently close approximant $\tilde{m}$ with values in the fixed range has a nondegenerate local minimum $d_{\tilde{m}} \in I$, where $M_{\tilde{m}}(d_{\tilde{m}})$ is close to, but in general not equal to, 1. (A sequential statement becomes the neighbourhood statement of the theorem by the usual argument by contradiction.) Each approximant is nonconstant, so Lemma 6 gives $M_{\tilde{m}} > \int \tilde{m}$ everywhere, with both endpoint limits equal to $\int \tilde{m}$, which is close to $\bar{m} < 1$ and hence below $M_{\tilde{m}}(d_{\tilde{m}})$. The argument of the previous paragraph, with 3 replaced by $d_{\tilde{m}}$, then gives two distinct strict local maxima. □

Proof of Proposition 4. The method is the one just described, with $d = 76/25$. Solving the 3×3 linear system in exact rational arithmetic gives $p_1, p_2, p_3 > 0$. Then $\sum p_i\theta_i = 1$ and each equilibrium equation hold exactly, so $M(d_0) = 1$ by the uniqueness in Lemma 5. The third equation is $M'(d_0) = 0$ by (15). The exact values of $D > 0$ and of the expectation in (17) give $M''(d_0) > 0$. The exact values, computed and asserted in rational arithmetic by `contrast_witness.py` and recorded in `contrast_witness.json`, are

$$p_3 = \frac{3570001575000175}{480662491691127955758767000019} \approx 7.43 \times 10^{-15},$$

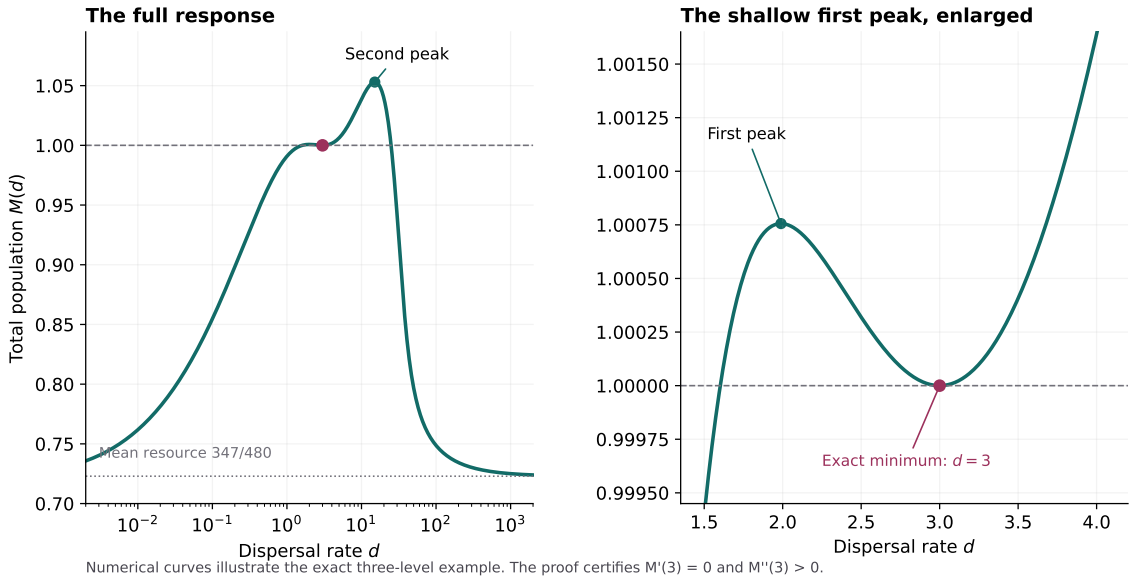


Figure 1: The exact three-level example of Theorem 2. Left: the full response on a logarithmic dispersal axis. Right: the shallow first maximum, enlarged. The minimum at $d = 3$ is certified by (4). The plotted curves and the marked maxima are computed numerically: the maxima by Brent's method on M', in the script that draws the figure. They illustrate the result but are not used in the proof.

$$D = \frac{549999325000133}{2221997272999905} \approx 0.2475,$$

with $p_1 \approx 0.0909092$, $p_2 \approx 0.9090908$, and $M''(d_0) = -2S/D$, where S is the expectation in (17), equals a positive rational with a 52-digit numerator and a 73-digit denominator, $M''(d_0) \approx 1.0797588 \times 10^{-21}$. Because the curvature is so small, we also give a check that does not use M''. By Lemma 5, the closure $F(d, M) = \mathbb{E}\Phi(m, d, M) - M$ is positive for $M < M(d)$ and negative for $M > M(d)$, so the sign of $F(d, 1)$ is the sign of $M(d) - 1$. Rigorous rational enclosures of the square roots in Φ (script `interval_traps.py`, standard library only) certify

$$M(d_0 - 10^{-4}) < 1, \quad M(d_0 - 10^{-5}) > 1, \quad M(d_0 + 10^{-2}) > 1, \quad M(\tfrac{31}{10}) < 1,$$

while $M(d_0) = 1$ exactly. So $(d_0 - 10^{-4}, d_0 - 10^{-5}, d_0)$ and $(d_0, d_0 + 10^{-2}, \frac{31}{10})$ are traps with disjoint open intervals, and Remark 8 gives two strict local maxima directly. Lemma 6 applies because the law is nonconstant. Every nonconstant law satisfies $M > \bar{m}$ and has endpoint limits $\bar{m}$, so the two-maxima argument in the proof of Theorem 2 applies without change. □

Effect sizes. Figure 1 shows how fragile the first example is. Numerically, the first maximum is at $d \approx 1.989$ with $M \approx 1.000756$. The certified minimum is at $d = 3$ with $M = 1$. The second maximum is at $d \approx 15.08$ with $M \approx 1.0530$. The first peak rises only about 7.6×10^{-4} above the minimum, a relative depth of about 0.08%. This is why the right panel enlarges it. The contrast is $m_3/m_1 = 658$, and the richest level occupies $1/876$ of the habitat. Table 1 compares the two exact examples. The low-contrast example of Proposition 4 buys its contrast with an even shallower dip: $M''(d_0)$ is of order 10^{-21}, and the rich patch has area about 7×10^{-15}. This example is quantitatively extremely fragile. Its left maximum lies within 10^{-4}

Table 1: Effect sizes of the exact examples (maxima located numerically; minima certified exactly).

	Theorem 2	Proposition 4
Contrast $\max m / \min m$	658	≈ 14.1734
Area of richest level	$1/876$	$\approx 7 \times 10^{-15}$
Certified minimum	$d = 3$, $M = 1$	$d = 3.04$, $M = 1$
M'' at the minimum	$\approx 3.06 \times 10^{-3}$	$\approx 1.08 \times 10^{-21}$
First maximum	$d \approx 1.989$, $M \approx 1.000756$	not resolved in double precision
Second maximum	$d \approx 15.08$, $M \approx 1.0530$	not resolved in double precision
Relative depth of first dip	$\approx 7.6 \times 10^{-4}$ (0.08%)	not resolved in double precision

of d_0 and rises only about 10^{-32} above $M(d_0) = 1$; its right maximum lies in $(d_0 + 10^{-2}, 3.1)$. The witness sits at the edge of the feasible three-level region: changing d_0 to 3.03, θ_3 to 11.3, θ_2 to $1 + 10^{-6}$ or θ_1 to $1 - 10^{-7}$ makes $M''(d_0)$ negative, and the two maxima of this construction disappear. The existence claim is nevertheless rigorous, because the certificate is exact. Both examples are qualitative. They show that multiple maxima exist, not that they would be visible in a census.

5 The response to rare resource patches

The idea in brief. Start from a constant habitat with resource a. Replace a tiny fraction εw of it with very rich habitat of resource r. The mean resource barely changes, but the population gains an extra amount $\varepsilon w\, g_r(d)$, where g_r is an explicit positive bump in the dispersal rate. It vanishes at very small and very large d and peaks near $d \approx r/2$ (Lemma 11). Now use several rich levels $r_1 \ll r_2 \ll \cdots \ll r_N$, each rarer than the last. The bumps sit at widely separated dispersal rates and barely interfere, so their sum has N separate humps (Figure 2). For small ε the true population is the mean resource plus ε times this sum, up to an error of order ε^2, so it inherits the N humps. Smoothing and rescaling then give Theorem 1.

Fix a background resource $a > 0$. For a resource level $r \neq a$, define

$$t_r(d) = \Phi(r, d, a), \qquad g_r(d) = \frac{d(t_r(d) - a)^2}{a\, t_r(d)}. \tag{18}$$

Here $t_r(d)$ is the density a site of resource r would reach if the pool were held at the background level a. The kernel g_r is the first-order population gain above the mean resource when a small part of a constant habitat is replaced by resource r.

Lemma 10 (First variation in patch abundance)**.** *Let $r_1, \dots, r_k \neq a$ be positive and $w_1, \dots, w_k > 0$, and put*

$$W = \sum_{i=1}^{k} w_i, \qquad \mu_\varepsilon = (1 - \varepsilon W)\delta_a + \varepsilon \sum_{i=1}^{k} w_i \delta_{r_i}, \quad 0 \le \varepsilon < W^{-1}.$$

Let $a_\varepsilon = \int z \, \mathrm{d}\mu_\varepsilon(z)$. For every compact interval $I = [d_0, d_1] \subset (0, \infty)$ there are $\varepsilon_I > 0$ and $C_I < \infty$ such that

$$\left| M_{\mu_\varepsilon}(d) - a_\varepsilon - \varepsilon \sum_{i=1}^{k} w_i g_{r_i}(d) \right| \le C_I \varepsilon^2 \qquad (d \in I,\ 0 \le \varepsilon \le \varepsilon_I). \tag{19}$$

Proof. For this finite atomic law the closure is

$$F(\varepsilon, d, M) = (1 - \varepsilon W)\Phi(a, d, M) + \varepsilon \sum_i w_i \Phi(r_i, d, M) - M = 0,$$

which is jointly real analytic in (ε, d, M) on a neighbourhood of $\{0\} \times I \times \{a\}$, even for small negative ε. At $\varepsilon = 0$ the root is $M = a$, and $F_M = -D = -a/(a+d) \le -a/(a+d_1) < 0$ uniformly on I. The analytic implicit function theorem gives, near each $d \in I$, an analytic root $M(\varepsilon, d)$. By continuity and compactness of I there are $\delta > 0$ and $\varepsilon_I \in (0, 1/(2W))$ such that $F_M \le -a/(2(a+d_1)) < 0$ whenever $|M - a| \le \delta$, $|\varepsilon| \le \varepsilon_I$ and $d \in I$. On this set F is strictly decreasing in M, so it has at most one root there. The local implicit-function solutions therefore agree on overlaps, and together they define one analytic function $M(\varepsilon, d)$ on $[-\varepsilon_I, \varepsilon_I] \times I$. The choice $\varepsilon_I < 1/(2W)$ keeps $1 - \varepsilon W > 0$, so μ_ε is a probability law for $0 \le \varepsilon \le \varepsilon_I$. For $\varepsilon \ge 0$ it coincides with $M_{\mu_\varepsilon}(d)$ by the uniqueness in Lemma 5. Differentiating $F = 0$ in ε at $\varepsilon = 0$, using $\Phi(a, d, a) = a$ and $\Phi_M = d/(a+d)$ there, gives

$$\left.\frac{\partial M}{\partial \varepsilon}\right|_{\varepsilon=0} = \frac{a+d}{a} \sum_i w_i (t_{r_i}(d) - a).$$

Since $a_\varepsilon = a + \varepsilon \sum_i w_i (r_i - a)$, the coefficient of ε in $M - a_\varepsilon$ is the weighted sum of $\frac{a+d}{a}(t_r - a) - (r - a)$. Using $r = t_r + d - da/t_r$, this equals $d(t_r - a)(1/a - 1/t_r) = g_r(d)$. Taylor's theorem in ε, with $C_I = \frac{1}{2} \max |\partial_\varepsilon^2 M|$ over the compact set $[0, \varepsilon_I] \times I$, gives (19) uniformly in $d \in I$. □

Lemma 11 (Separated scales)**.** *For each fixed $r \ne a$, $g_r(d) > 0$ for $d > 0$, and*

$$\lim_{d \downarrow 0} g_r(d) = \lim_{d \to \infty} g_r(d) = 0.$$

As $R \to \infty$,

$$R^{-2} g_R(d) = O(R^{-1}) \longrightarrow 0 \qquad \text{for every fixed } d > 0, \tag{20}$$

$$R^{-2} g_R(sR) \longrightarrow \frac{s(1-s)}{a} \qquad \text{for } 0 < s < 1, \tag{21}$$

$$R^{-2} g_R(R) \longrightarrow 0. \tag{22}$$

Proof. The root t_r lies strictly between a and r, so positivity is immediate. The first endpoint limit follows from $t_r(0) = r$. For $d \to \infty$, the equilibrium identity gives $t_r - a = t_r(r - t_r)/d = O(d^{-1})$, and hence $g_r(d) = O(d^{-1})$. For fixed d, $t_R(d)/R \to 1$, so $g_R(d) = O(R)$. For $0 < s < 1$, the quadratic formula gives

$$\frac{t_R(sR)}{R} = \frac{1 - s + \sqrt{(1-s)^2 + 4sa/R}}{2} \longrightarrow 1 - s.$$

Substitution into (18) proves (21). Finally $t_R(R) = \sqrt{aR}$, and substitution proves (22). □

Lemma 12 (Positive sums with any prescribed number of humps)**.** *For every $N \ge 1$, there are*

$$a < r_1 < \cdots < r_N, \qquad r_{k+1} > 2r_k,$$

and N pairwise disjoint intervals $[\alpha_j, \gamma_j] \subset (0, \infty)$, containing points $\alpha_j < \beta_j < \gamma_j$, for which the function

$$G_N(d) = \sum_{i=1}^{N} r_i^{-2} g_{r_i}(d) \tag{23}$$

satisfies

$$G_N(\beta_j) > \max\{G_N(\alpha_j), G_N(\gamma_j)\}, \qquad j = 1, \ldots, N. \tag{24}$$

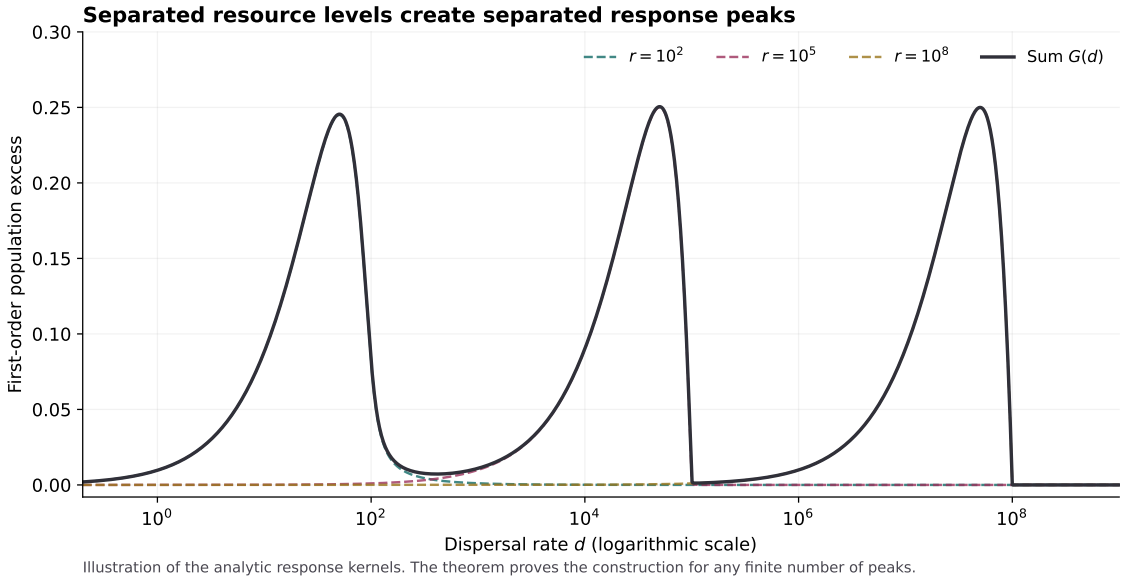


Figure 2: The first-variation construction with background $a = 1$, resource levels $10^2, 10^5, 10^8$ and weights r_i^{-2}. The curve shown is the kernel sum G_N, not the nonlinear population M_{μ_ε}. The true population excess is ε times this curve, so its vertical scale is small. Lemma 10 transfers strict inequalities from this sum to the nonlinear equilibrium for all sufficiently small $\varepsilon > 0$.

Proof. We construct the resource levels inductively, starting with $G_0 = 0$ and no intervals, with the convention $r_0 := a$. Suppose G_k and its k strict trapping inequalities have been chosen. Since $G_k(d) \to 0$ as $d \to \infty$, choose $L > r_k$ larger than all right endpoints chosen so far, with $G_k(L) < 1/(16a)$. For R larger than all chosen resource levels, define

$$G_{k+1}(d) = G_k(d) + R^{-2} g_R(d).$$

At each of the finitely many old trapping points, the added term tends to zero by (20), so the old inequalities survive for all sufficiently large R. Moreover, using Lemma 11 and the decay at infinity of the finitely many old kernels,

$$G_{k+1}(L) \longrightarrow G_k(L) < \frac{1}{16a}, \qquad G_{k+1}(R/2) \longrightarrow \frac{1}{4a}, \qquad G_{k+1}(R) \longrightarrow 0.$$

Hence, for all sufficiently large $R > 2L$, we have $G_{k+1}(L) < 1/(8a)$, $G_{k+1}(R/2) > 3/(16a)$ and $G_{k+1}(R) < 1/(16a)$. So the triple $(L, R/2, R)$ is a new strict trap, disjoint from all the old ones. Set $r_{k+1} = R$, which exceeds $2L > 2r_k$, and continue. □

Proof of Theorem 1. Step 1: atomic laws. Take the levels and the $3N$ trapping points from Lemma 12, and let I be a compact interval containing all of them. Let $\Delta > 0$ be the smallest margin in (24). Apply Lemma 10 on I with $w_i = r_i^{-2}$. Since a_ε does not depend on d, every difference $M_{\mu_\varepsilon}(\beta_j) - M_{\mu_\varepsilon}(\alpha_j)$ equals $\varepsilon(G_N(\beta_j) - G_N(\alpha_j))$ up to $2C_I\varepsilon^2$, and similarly at γ_j. So for $0 < \varepsilon < \varepsilon_* := \min\{\varepsilon_I, \Delta/(4C_I)\}$, each of the N traps holds for M_{μ_ε} with margin at least $\varepsilon\Delta/2$. The traps lie in disjoint intervals, so Remark 8 gives N distinct strict local maxima. The law μ_ε has $N + 1$ levels, including the background a.

Step 2: smoothing, with the tolerance chosen after ε. Realise μ_ε as the nondecreasing step function s_ε on $(0, 1)$, with the background first and the other levels in increasing order. Its

range lies in $[a, r_N]$, and $\|s_\varepsilon - a\|_{L^1} = \varepsilon C_N$ with $C_N = \sum_i r_i^{-2}(r_i - a)$. Fix $\varepsilon \in (0, \varepsilon_*)$. By Lemma 7, applied with the common range $[a, r_N + 1]$, there is $\eta_\varepsilon > 0$ such that every resource v with values in $[a, r_N + 1]$ and $\|v - s_\varepsilon\|_{L^1} < \eta_\varepsilon$ still satisfies all N trapping inequalities. The mollification construction in the proof of Theorem 2 gives a smooth v_ε with

$$v_\varepsilon' > 0, \quad a \le v_\varepsilon \le r_N + 1, \quad \|v_\varepsilon - s_\varepsilon\|_{L^1} < \min\{\varepsilon^2, \eta_\varepsilon\}.$$

Both bounds hold at once because the added linear term κx and the mollifier width can be taken as small as we like. A mollifier width below the length of the top plateau also keeps $\max v_\varepsilon \ge r_N$.

Step 3: normalisation. Put $b_\varepsilon = \int_0^1 v_\varepsilon$. Then $|b_\varepsilon - a| \le \varepsilon C_N + \varepsilon^2$, so $c_\varepsilon := a/b_\varepsilon \to 1$. Set $m_\varepsilon = c_\varepsilon v_\varepsilon$. It has total resource exactly a, and it is positive and strictly increasing. By the scaling law (11) it has the same number of strict local maxima as v_ε. Since $a \le v_\varepsilon \le a + \kappa$ near $x = 0$ and $\max v_\varepsilon \ge r_N$, its contrast is at least $r_N/(a + \kappa)$. This exceeds 2^{N-1} once κ is small, because $r_N > 2^{N-1} r_1 > 2^{N-1} a$.

Step 4: convergence. We have $\|m_\varepsilon - a\|_{L^1} \le c_\varepsilon \|v_\varepsilon - s_\varepsilon\|_{L^1} + c_\varepsilon \|s_\varepsilon - a\|_{L^1} + |c_\varepsilon - 1| a \to 0$. The functions are uniformly bounded by $2(r_N + 1)$ for small ε, so for each finite $p \ge 1$,

$$\|m_\varepsilon - a\|_{L^p}^p \le \|m_\varepsilon - a\|_{L^\infty}^{p-1} \|m_\varepsilon - a\|_{L^1} \longrightarrow 0.$$

The convergence is not uniform: $\max m_\varepsilon \ge c_\varepsilon r_N$ with $c_\varepsilon \to 1$, so $\liminf_{\varepsilon \downarrow 0} \|m_\varepsilon - a\|_{L^\infty} \ge r_N - a > 0$. Taking any sequence $\varepsilon_j \downarrow 0$ in $(0, \varepsilon_*)$ and $m_j = m_{\varepsilon_j}$ completes the proof. □

6 Ecological reading

Model (1) describes a species whose dispersers enter a well-mixed pool and settle at random over the whole habitat. Examples include marine species with a well-mixed larval pool (the open-recruitment setting of fisheries), wind-dispersed seeds or spores, and fully mixed laboratory metapopulations. The resource m acts as the local carrying capacity, and the dispersal rate d measures connectivity. For rational habitat fractions the model is exactly a fully connected network of equal patches (Section 7). Within these assumptions, the results suggest four qualitative points.

1. *Responses to connectivity need not be monotone, or even single-peaked.* Raising connectivity (corridors, removing barriers) or lowering it (fragmentation) can raise or lower equilibrium abundance, and can do so more than once. Results at one or two connectivity levels cannot safely be extrapolated.

2. *Identifiability.* With several peaks, the same total abundance occurs at several dispersal rates. Total stock size alone cannot then identify connectivity.

3. *Rare rich patches matter.* In the mechanism of Section 5, very small, very productive patches (spawning or nursery grounds, upwelling cells, refugia) control the response at their own dispersal scale while being nearly invisible in the mean resource.

4. *A simple rule for binary habitats.* If habitat is effectively good or poor, Theorem 3 predicts a single abundance peak at $d = (a + b)/2$, with the explicit height (6). This is a testable prediction.

These points are qualitative. Our examples need strong contrast, and their extra peaks are far too shallow to see in a census (Table 1). The model also omits dispersal mortality, Allee effects, temporal variability and harvesting. Whether peaks of ecologically visible depth can occur under moderate contrast is open (Section 8).

7 Related work

Total population versus dispersal. Lou [13] and DeAngelis, Ni and Zhang [7] study how migration and spatial heterogeneity affect the total population of a single species. This line of work includes the observation that total population can exceed total resource, which for our model is Lemma 6. For random diffusion, Liang and Lou [11] study how population size depends on the dispersal rate. Bai and Li describe that work as showing several maxima for near-constant resources under local diffusion, a different operator. Bai and Li [3] posed our question for the uniform nonlocal operator. Bai, Li and Zhou [5] optimise total population when the resource distribution and the dispersal rate vary together. They also concentrate resources on small sets, but they study a different quantity from the number of maxima along the dispersal curve of a fixed resource.

Patch models. Freedman and Waltman [9] studied two coupled habitats, including the perfect-mixing limit of the total. Arditi, Lobry and Sari [1] analyse the standard two-patch logistic model in its full parameter space. The total may be above or below the sum of carrying capacities, and in one domain it is above at low migration and below at high migration. They also show that mechanistic consumer–resource couplings can give different predictions; one such two-patch curve first decreases, then increases, then decreases. They leave more than two patches as a challenge. Their 2018 paper [2] treats asymmetric migration between two patches. Each of its total-population curves either rises to a single maximum and then falls, or stays on one side of the sum of carrying capacities. We read both papers in full. Neither contains a curve with more than one interior local maximum, and neither treats the complete-graph model with three or more patches. Gao and Lou [10] classify total biomass in the general two-patch model. Our Theorem 3 is a short self-contained proof of the specialisation needed here, not a new classification. Elbetch, Benzekri, Massart and Sari [8] study the multi-patch logistic equation with asymmetric migration. Their three-patch examples show complex responses, including several crossings of the sum-of-carrying-capacities baseline. Their migration matrices and growth-to-competition ratios are general. Our population always lies strictly above the mean resource (Lemma 6), so those crossing examples do not carry over unchanged. Segura, Marvá and Franco [15] give numerical three-patch scenarios with independently variable growth and competition coefficients. Bravo de la Parra [6] studies dispersal and total population size in discrete-time metapopulation models. Our claim is specific to (1): a common quadratic coefficient, a fixed uniform mixing operator, and variation of its single scalar rate.

The identical-migration patch model. Liang and Zhang [12] study the n-patch logistic model with identical migration. With n equal patches, $\dot{u}_i = d(\frac{1}{n}\sum_j u_j - u_i) + u_i(m_i - u_i)$, this is the equal-patch discrete analogue of (1). Every resource law with rational weights $p_i = k_i/n$ is realised in it by giving k_i patches the value m_i. In particular, Theorem 2 is an example with $n = 3504$ patches. The three-level examples therefore live inside Liang and Zhang's model, which makes it the closest source of priority risk. Only its abstract was available to us. The abstract concerns the joint optimisation of resource distribution and migration rate, and does not report multiple maxima, but we cannot exclude that the full text discusses them.

Follow-up by the source authors. Bai, Li and Shi [4] survey single-species models with nonlocal dispersal in heterogeneous environments. Its abstract announces a new monotonicity result for the total population and does not report multiple maxima. We found no subsequent

resolution in the sources we could inspect; the full text of [4] was not available to us. Only its abstract was read.

8 Open problems

The contrast threshold. Let κ_3 be the infimum of $\max m / \min m$ over three-level laws whose population curve has at least two strict local maxima. Bai and Li's theorem and Proposition 4 give

$$\frac{1+\sqrt{5}}{2} \;\le\; \kappa_3 \;\le\; 14.1734.$$

We have not proved any better lower bound. Two numerical computations suggest that the upper bound is nearly sharp, but they are evidence only, not proofs. First, a multistart constrained optimisation over all three-level laws, parametrised as in Section 4, found no witness below 14.17; this is weak evidence, because its best contrast was 14.41 and it never reached the certified 14.17 itself. Its iterates drift towards the rare-patch limit, in which the background level and a second level merge and the rich level becomes vanishingly rare. Second, in that limit, multiple maxima reduce to two humps of $c\,d/(1+d)^2 + g_R(d)$ with $a = 1$. This first appears at $R^* \approx 14.173$. Proving $\kappa_3 = R^*$, or any lower bound above 1.618, is open. So is the analogous threshold for $k \ge 4$ levels, and for arbitrary laws.

Contrast needed for N maxima. Our construction uses contrast above 2^{N-1}. We do not know whether N maxima force a contrast growing with N. A bound of the form "contrast below κ allows at most $N(\kappa)$ maxima" would extend Bai and Li's theorem, which is the case $N = 1$.

Depth and levels. Can peaks of non-negligible relative depth (say 5%) occur under bounded contrast? Are $N + 1$ levels necessary for N maxima when $N \ge 3$? Does the rare-patch mechanism persist for convolution kernels close to uniform mixing, for non-complete patch graphs, or with dispersal mortality?

Declarations

Author. Anonymous.

AI provenance. The mathematics, computations and text of this paper were produced with substantial assistance from AI systems, and revised after simulated multi-reviewer assessment. All load-bearing identities are verified by exact rational arithmetic in the accompanying scripts.

Literature caveats. The originating question was checked in the full text of [3], and both papers of Arditi, Lobry and Sari were read in full. For Liang and Zhang [12] and for Bai, Li and Shi [4], only the abstracts were read, and neither abstract reports multiple maxima. Liang and Zhang's model is the equal-patch discrete analogue of (1) and contains our rational examples, so it carries a real priority risk. We found no subsequent resolution in the sources we could inspect; the full text of [4] was not available to us.

Code and data availability. The scripts, receipts and figures are archived at `https://doi.org/10.5281/zenodo.23687852` and in `https://github.com/ipitchford/multiple-dispersal-maxima` (tag `v1.0.0-candidate`). The accompanying scripts write to an output directory (default `output/`) and never overwrite the shipped receipts. `audit_meanfield.py` verifies Theorem 2 exactly. `contrast_witness.py` verifies Proposition 4 and Theorem 2 exactly. `resource_fingerprints.py` generates exact instances and computes all matrix invariants. These three use only the Python standard library. `plot_population.py` draws the figures and computes the plotted maxima. `contrast_search.py` runs the numerical threshold search of Section 8. `interval_traps.py` certifies the two traps in the proof of Proposition 4, using only the standard library. The standard-library scripts run with `uv run python3 -I`. The other two declare their pinned dependencies inline (PEP 723) and in `requirements.txt`; run them with `uv run --with numpy==2.4.6 --with scipy==1.17.1 --with matplotlib==3.11.2 python3 -I`. All were rerun from a clean copy under Python 3.11 with these versions, which are recorded in `environment.json`. Figures are not byte-reproducible across platforms. The multistart search of Section 8 is also platform-sensitive: its reported best contrast 14.41 was obtained on macOS (arm64); two Linux (x86-64) runs with the same pinned versions found no successful start and 14.1745 respectively, neither below the certified 14.1734. The rare-patch threshold R^* agrees across platforms. The proofs use strict inequalities and exact identities and do not depend on plotting resolution.

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